ERICT-PE-11_YEL-QoS_Congestion_Event_SOP

Node: YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_8201

Description

The QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain on the Ericsson vendor equipment PE (Packet Engine) role at the Yelahanka Sub-station site, node ID YEL-PE-11, is a critical issue that arises due to the mismanagement of packet flow and bandwidth allocation mechanisms. The PE node is responsible for processing, managing, and forwarding packet traffic, while adhering to the predefined quality of service (QoS) policies. The congestion event can be triggered by microbursts, which are temporary, high-intensity packet bursts that overwhelm the available bandwidth, causing interface tail drops and subsequent packet loss. This, in turn, leads to QoS queue starvation, where critical packets are delayed or dropped, compromising the overall network performance and user experience. To mitigate this issue, it is essential to investigate the root cause of the congestion event, which could be attributed to configuration errors, hardware malfunctions, or inadequate network resource allocation.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Yelahanka Sub-station site to physically verify the node and network infrastructure

- Verify the status of redundant links and ensure they are functioning as expected

- Review recent configuration changes and logs to identify potential culprits

Potential Root Causes

Cause	Probability
Misconfigured QoS policies or bandwidth allocation	High
Hardware malfunction or interface issues	Medium
Insufficient network resources or inadequate resource allocation	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy | include microburst, Expected Output: Microburst detection enabled on all interfaces, Description: This command checks if microburst detection is enabled on all interfaces, helping to identify potential causes of congestion events

Command: show interface | include tail-drop, Expected Output: Tail drop events: 10, Description: This command verifies the number of tail drop events on each interface, indicating potential congestion

Command: show qos queue | include starved, Expected Output: QoS queue starvation detected on queue 1, Description: This command identifies QoS queue starvation, which can lead to critical packet drops
```

Resolution Steps

Step 1: Step: Verify and correct QoS policy configuration, Command: config qos policy | set microburst-detection enabled, Description: This step involves reviewing and adjusting QoS policies to ensure optimal packet flow and bandwidth allocation

Step 2: Step: Replace or repair malfunctioning hardware, Command: hw replace interface , Description: This step requires replacing or repairing faulty hardware to prevent congestion events

Step 3: Step: Allocate additional network resources, Command: config resource | set allocation , Description: This step involves allocating additional resources to prevent QoS queue starvation

Post-Check Verification

```
Command: show qos policy | include microburst, Expected Output: Microburst detection disabled on all interfaces, Description: This command verifies that microburst detection is disabled on all interfaces, ensuring optimal packet flow

Command: show interface | include tail-drop, Expected Output: Tail drop events: 0, Description: This command confirms that there are no tail drop events on any interface, indicating resolved congestion

Command: show qos queue | include starved, Expected Output: No QoS queue starvation detected, Description: This command verifies that QoS queue starvation has been eliminated, ensuring critical packets are processed without delay
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: config qos policy | set microburst-detection , Description: This step involves restoring the original QoS policy configuration to prevent unintended consequences

- Step: Revert hardware changes, Command: hw revert interface , Description: This step requires reverting any hardware changes made during the resolution process to prevent service disruptions

- Step: Revert resource allocation changes, Command: config resource | set allocation , Description: This step involves reverting any resource allocation changes made during the resolution process to prevent service disruptions